African Institute for Mathematical Sciences RWANDA

Introduction to Probability and Statistics

Lecture 10: Convergence of the mean, Central Limit Theorem

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1. The statistic $\overline{X}_n$

2. Central Limit Theorem

WLLN

Weak Law of Large Number (WLLN)

Let $X_1, \cdots, X_n$ be n independent and identically distributed (iid) r.v.'s such that $E(X_i) = a < +\infty$ and $Var(X_i) = b$, $i = 1, \ldots, n$.
Then $\overline{X}_n$ **converge in probability** to a.

Proof: Apply Chebyshev's inequality

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$$\forall \varepsilon > 0, P(|\overline{X}_n - a| > \varepsilon) \leqslant \frac{b}{n\ \varepsilon^2} \underset{n\to+\infty}{\longrightarrow} 0$$

$$\iff \quad \forall \varepsilon > 0, \lim_{n\to+\infty} P(|\overline{X}_n - a| > \varepsilon) = 0$$

$$\iff \quad \overline{X}_n \xrightarrow{P} a.$$

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  where $X \sim \delta_a$, the **point mass** (or **Dirac**) **distribution**.

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- $\Longrightarrow$ Convergence in Probability: $\Longrightarrow \quad \overline{X}_n \xrightarrow{P} a$.

Limit of $\overline{X}_n$ m.g.f

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$$M_{\overline{X}_n}(t) = E\left(e^{t\overline{X}_n}\right) = E\left(e^{\frac{t}{n}\sum_{i=1}^n X_i}\right) = \prod_{i=1}^{n} E\left(e^{\frac{t}{n}X_i}\right) = \left[E\left(e^{\frac{t}{n}X_i}\right)\right]^n$$

But $e^{\frac{t}{n}X_i} = 1 + \frac{t}{n}X_i + \frac{t^2}{n^2}X_i^2 + \cdots$

Then $E\left(e^{\frac{t}{n}X_i}\right) \sim \left[1 + \frac{t}{n}E(X_i) + o\left(\frac{t}{n}\right)\right]^n = \left[1 + \frac{ta}{n} + o\left(\frac{t}{n}\right)\right]^n$

where o is a function such that $\lim_{h\to 0} o(h)/h = 0$.

$$\begin{aligned}
\left[1 + \frac{ta}{n} + o\left(\frac{t}{n}\right)\right]^n &= \exp\left\{n \log\left[1 + \frac{at + n\ o(t/n)}{n}\right]\right\} \\
&= \exp\left\{n \times \frac{at + n\ o(t/n)}{n}\right\} \\
&= \exp\left\{at + n\ o(t/n)\right\}
\end{aligned}$$

which goes to $\exp\{at\}$ when n goes to infinity since $n\ o(t/n)$ goes to 0.
Hence

$$M_{\overline{X}_n}(t) \underset{n\to+\infty}{\longrightarrow} e^{ta}.$$

Convergence in distribution

$$\Longrightarrow \quad \overline{\mathbf{X}}_{\mathbf{n}} \xrightarrow{\mathbf{d}} \mathbf{X}, \quad \textit{where } X \sim \delta_a.$$

Convergence in distribution

Z_n converges **in distribution** or **in law** if,

$$\lim_{n \to +\infty} F_{Z_n}(z) = F_Z(z),$$

at each point z where F_Z is continuous.

Notation: $Z_n \xrightarrow{d} Z$ or $Z_n \xrightarrow{\mathscr{L}} Z$

Point mass distribution

$$\implies \quad \overline{X}_n \xrightarrow{d} X, \text{ where } \mathbf{X} \sim \delta_{\mathbf{a}}.$$

Point mass distribution

A r.v. X on $\mathbb{R}$, is said to follow the **point mass distribution** δ_c, if $P(X = c) = 1$.
The c.d.f of X is defined as:

$$F(x) = \begin{cases} 0 & x < c \\ 1 & x \geqslant c \end{cases}$$

Moment generating function:

$$M_X(t) = E(e^{tX}) = e^{tc} P(X = c) = e^{tc}.$$

Convergence in distribution $\overline{X}_n$

Continuity Theorem for m.g.f.'s

Let $Z_1, Z_2, \ldots$ be a sequence of r.v.'s, F_n be the c.d.f of Z_n and $M_{Z_n}(t)$ the m.g.f of Z_n. Then:

$$\left(M_{Z_n}(t) \underset{n\to+\infty}{\longrightarrow} M_Z(t)\right) \Longleftrightarrow \left(Z_n \xrightarrow{d} Z\right).$$

Proposition

Let $X_1, X_2, \ldots$ be a sequence of r.v.'s, with F_n, the c.d.f of X_n. Let X be a r.v with the point mass distribution δ_c. Then:

$$\left(X_n \xrightarrow{d} X\right) \Longleftrightarrow \left(X_n \xrightarrow{P} c\right).$$

Proof of the proposition

- Suppose $\lim_{n\to+\infty} F_{X_n}(x) = F_X(x)$ with $X \sim \delta_c$.

$$\begin{aligned}\forall\varepsilon > 0, P(|X_n - c| > \varepsilon) &= P(X_n > c-\varepsilon) + P(X_n < c+\varepsilon)\\ &= 1 - F_n(c-\varepsilon) + F_n(c-\varepsilon)\end{aligned}$$

$\lim_{n\to+\infty} F_{X_n}(c-\varepsilon) = F_X(c-\varepsilon) = 0$ and $\lim_{n\to+\infty} F_{X_n}(c+\varepsilon) = F_X(c+\varepsilon) = 1$.

Hence $X_n \xrightarrow{P} c$.

- Suppose that $\forall\varepsilon > 0,\ P(|X_n - c| > \varepsilon) \to 0$ when $n \to +\infty$.
We have $P(X_n > c+\varepsilon) + P(X_n < c-\varepsilon) \longrightarrow 0$.
Therefore $P(X_n < c-\varepsilon) = F_n(c-\varepsilon) \longrightarrow 0$ and $\lim F_n(x) = 0,\ \forall x < c-\varepsilon$ (since $F_n \uparrow$).
In the same way $P(X_n > c+\varepsilon) = 1 - F_n(c+\varepsilon) \longrightarrow 0$ and $\lim F_n(x) = 1,\ \forall x > c+\varepsilon$.
Hence $\lim F_n(x) = F_X(x)$ such that $F_X(x) = 1$ for $x \geqslant c$ and $F_X(x) = 0$ for $x > c$ which means that X_n converges to X having a point mass probability δ_c.

CLT

Central Limit Theorem (CLT)

Let $X_1, \cdots, X_n$ be n independent and identically distributed (iid) r.v.'s with $E(X_1) = \mu$ and $Var(X_1) = \sigma^2 > 0$ (both finite).
Then:

$$\sqrt{n}\left(\frac{\overline{X}_n - \mu}{\sigma}\right) \sim \mathcal{N}(0,1) \text{ when } n \to +\infty.$$

The r.v. $Z_n = \sqrt{n}(\overline{X}_n - \mu)/\sigma$ converges **in distribution** (or in law) to a standard normal r.v. or X_n behaves as a normal distribution with parameters $(\mu, \sigma^2/n)$ when n goes to infinity.

$$\lim_{n \to +\infty} F_{Z_n}(z) = \Phi(z)$$

.

CLT

Proof: We are going to show that $M_{Z_n}(t) \underset{n\to+\infty}{\longrightarrow} M_Z(t)$.

$Z_n = \sqrt{n}\left(\frac{\overline{X}_n-\mu}{\sigma}\right) = \frac{1}{\sqrt{n}}\sum_{i=1}^n\left(\frac{X_i-\mu}{\sigma}\right)$

Let $Y_i = \dfrac{X_i-\mu}{\sigma}$, $Z_n = \dfrac{1}{\sqrt{n}}\displaystyle\sum_{i=1}^n Y_i$.

$M_{Z_n}(t) = E(e^{tZ_n}) = \prod_{i=1}^n E(e^{\frac{t}{\sqrt{n}}Y_i}) = \left[E(e^{\frac{t}{\sqrt{n}}Y_i})\right]^n.$

$e^{\frac{t}{\sqrt{n}}Y_i} \approx 1 + \frac{t}{\sqrt{n}}Y_i + \frac{t^2}{2n}Y_i^2 \iff E(e^{\frac{t}{\sqrt{n}}Y_i}) \approx 1 + \frac{t}{\sqrt{n}}E(Y_i) + \frac{t^2}{2n}E(Y_i^2)$

Then $E(e^{\frac{t}{\sqrt{n}}Y_i}) \approx 1 + \frac{t^2}{2n}$ since $E(Y_i) = 0$ and $Var(Y_i) = E(Y_i^2) = 1$.

$\left[E(e^{\frac{t}{\sqrt{n}}Y_i})\right]^n \approx (1 + \frac{t^2}{2n})^n.$

But $(1 + \frac{t^2}{n})^n = e^{nlog(1+\frac{t^2}{2n})} \approx e^{\frac{t^2}{2}}$ since $log(1 + \frac{t^2}{2n}) \approx \frac{t^2}{2n}$ when n large.

Therefore $M_{Z_n}(t) \approx e^{\frac{t^2}{2}}$ which is the m.g.f. of a standard normal distribution.

CLT

Example:

Let $X_1, \cdots, X_n$ be independent Bernoulli r.v.'s with parameter p.
We have $E(X_i) = p$ and $Var(X_i) = p(1-p)$.
Applying the CLT leads to :

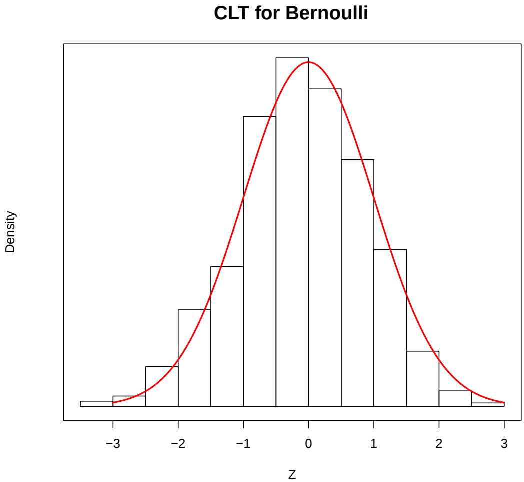
$$\sqrt{n}\left(\frac{\overline{X}_n - p}{\sqrt{p(1-p)}}\right) \sim \mathcal{N}(0,1).$$

Simulation:

- Choose a value for p e.g. $p = 0.8$.
- Draw 100 realizations of a Bernoulli with parameter 0.8.
- Compute $\overline{X}_n$, the mean of the 100 values.
- Reproduce this experiment 1000 times to obtain a sample of $\overline{X}_n$.
- Build the histogram to investigate the distribution of the variables

$$\sqrt{n}\left((\overline{X}_n - p)/\sqrt{p(1-p)}\right).$$

Illustration of the CLT



R code

```
p<-0.8 # Choose a value for p
n<-100  # Size of the sample
# Generate a sample of n Bernoulli r.v.'s realizations
x<-sample(c(0,1), n, replace=TRUE, prob=c(0.2,0.8))
M<-1000  # Number of samples
# Z = sqrt(n)(X̄_n - p)/sqrt(p(1 - p))
Z<-rep(0,M) # Initialization
for (i in 1:M)
Z[i]<-sqrt(n)*((mean(rbinom(n,1,p))-p)/sqrt(p*(1-p)))
# Histogram
hist(Z, nclass=20, probability=T, main=list("CLT for
Bernoulli",cex=1.5),
ylab="Density",yaxt = "n")
# Adding the p.d.f.
z<-seq(-3,3,0.01)
gaussz<-dnorm(z,0,1)
lines(z,gaussz)
box() # To add a frame
```